Felipe Vaz Peres

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EXPERIENCE

Huna **Remote**
Data Scientist *Sep 2025 - Present*
- Develop interpretable risk-stratification models for early cancer detection, prioritizing coefficient-level transparency and clinical auditability
- Co-authored an ASCO 2026 abstract on prostate cancer risk stratification in 2,978 men across seven Brazilian diagnostic networks, achieving 0.77 AUC and 35.1% fewer false positives versus PSA ≥4 ng/mL

Center for Microbial Secondary Metabolites, Technical University of Denmark **Remote**
Guest Researcher *Jan 2024 - Jan 2025*
- Developed an end-to-end long-read metagenomics pipeline for 100+ marine samples, generating 40 high-quality microbial genomes (>90% completeness)
- Applied ML-based contig classification to distinguish prokaryotic and eukaryotic sequences, enabling genome reconstruction from complex microbial communities
- Standardized biosynthetic gene cluster annotation across 40 genomes, cataloging 148 full-length BGCs
- Automated analysis and benchmarking workflows, reducing manual execution and improving reproducibility

Computational, Evolutionary and Systems Biology Laboratory, University of São Paulo **Piracicaba, SP**
Graduate Researcher *Jan 2022 - Jan 2025*
- Applied ML models (SVM, Random Forest, CNN) for large-scale ncRNA identification across 48 sugarcane genotypes, identifying 8.4M candidate ncRNAs
- Built and analyzed co-expression networks containing 1M+ nodes, identifying 7,000+ ncRNA-protein modules, revealing functional roles across diverse genetic backgrounds
- Functionally annotated 250K+ ncRNAs establishing a framework for investigating biological functions in this complex polyploid crop
- Contributed to 5 open-source projects across bacteria, microalgae, and plants, building automated Nextflow pipelines used by 3 research groups

Genera **Remote**
Research Intern *Jan 2020 - Mar 2020*
- Improved a human ancestry classification pipeline analyzing 650K+ SNPs across 72 population groups, including underrepresented South American populations
- Developed technical documentation for the ancestry inference workflow

Computational, Evolutionary and Systems Biology Laboratory, University of São Paulo **Piracicaba, SP**
Undergraduate Researcher *May 2019 - Jan 2022*
- Processed 3.7B sequencing reads (>1.1 TB) on HPC infrastructure to construct a sugarcane pan-transcriptome spanning 48 genotypes and 16M transcripts
- Developed an automated Snakemake pipeline for transcriptome assembly and comparative analysis of 600K+ protein variants
- Presented reproducible workflow design at ISMB, demonstrating scalability across polyploid crop species

EDUCATION

University of São Paulo, Center of Nuclear Energy in Agriculture **Piracicaba, SP**
M.Sc. in Bioinformatics *Jan 2022 - Jan 2025*
- Relevant coursework: Genomics, Bioinformatics

Federal University of São Carlos, Center of Agrarian Sciences **Araras, SP**
B.S. in Biotechnology *Mar 2018 - Jan 2022*
- Relevant coursework: Statistics, Bioinformatics, Molecular Biology, Genetic Engineering

TECHNICAL SKILLS

Programming: Python, Bash, R
Machine Learning: scikit-learn, statistical modeling, model validation, interpretable ML, MLflow
Clinical Risk Modeling: risk stratification, calibration, ROC/AUC, decision curve analysis, cross-validation, external validation, fairness and bias auditing
Computational Biology: metagenomics, genomics, transcriptomics, proteomics, NGS (short and long reads), gene expression, co-expression networks, functional annotation, clinical biomarkers, cohort data analysis
Data & Infrastructure: Databricks, GCP, Unix/Linux, HPC (SGE, PBS, LSF, SLURM), git

HONORS AND AWARDS

Liga Brasileira de Bioinformática
Third place *Jun 2025*
- Awarded third place at the largest bioinformatics competition in Latin America, solving complex computational biology challenges

National Council for Scientific and Technological Development (CNPq)
Graduate Research Fellowship *Apr 2022 - Apr 2024*
- Received a 24-month master's research fellowship funded by Brazil's national research council and awarded through the University of São Paulo's graduate program

Mendelics Challenge
Third place *Nov 2021*
- Awarded third place by developing an automated variant calling pipeline in under 48 hours using real genomic data

International Symposium of Undergraduate Research at University of São Paulo
Honorable mention *Oct 2020*
- Recognized for outstanding research activity at USP's 28th International Symposium of Undergraduate Research

São Paulo Research Foundation (FAPESP)
Undergraduate Research Fellowship *Feb 2020 - Jan 2022*
- Competitive undergraduate research fellowship awarded based on academic performance and research activity, with a merit-based renewal for a second 12-month term

Synthetic Biology Hackathon at Federal University of São Carlos
Second place *Aug 2018*
- Awarded second place by designing a synthetic biology bioremediation project presented at Brazil's largest biotechnology conference

OPEN-SOURCE PROJECTS

Additional work can be seen on my GitHub profile: https://github.com/felipevzps

Selected work
- **KAPT**, **R2C**: Nextflow pipelines for proteome inference and co-expression network construction, in active use by external research groups
- **seabed-symphony**: Pipeline for BGCs identification in microbial community of marine sediments

SELECTED PRESENTATIONS

2022 **Python: From zero to your first software**
Invited talk - Python Workshop for Biological Data, Online Conference - UNICAMP

An introduction to programming for bioscientists
Invited talk - Genetics and Molecular Biology Meeting, Online Conference - UNICAMP

2021 **Removing reads from contaminating organisms in NGS datasets**
Poster - Brazilian Symposium on Bioinformatics, Online Conference - Federal University of Minas Gerais